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Summary

Biosignal algorithm engineer working with EEG, ECG, PPG, respiration and motion data, from recording and quality control through to analysis and validated models. Master's thesis on decoding imagined movement from beta bursts in patients with tetraplegia, locked-in syndrome and disorders of consciousness. I build the evaluation harness before tuning anything, and I write down every preprocessing decision with the reason for it.

Experience

Dognosis **Bengaluru, India**
Research Associate *2026*

- Led research paradigms, collected and analysed the data, and designed the quality checks for the experiments.
- **Multi-stream acquisition and data quality:** Owned data-quality validation for a synchronised rig: four-channel EEG at 250 Hz, respiration at 475 Hz, nine-axis motion at 500 Hz, event timing, sample provenance and video, all on one LSL clock and recorded to XDF. Found three quality metrics that had silently stopped working, on three different signals, and added checks for each.
- **Signal processing:** Built an automated ECG-detection pipeline for EEG electrode channel verification: amplitude-independent beat detection from inter-peak-interval histograms, channel-relative artifact rejection and a contextual-confidence stage.
- **Validation:** Built the supervised evaluation harness before tuning: hand-labelled windows, per-second F1 with a grace margin, and leave-one-recording-out cross-validation.
- **Data engineering:** Root-caused a validation regression to a single commit and shipped the merged fix. Redesigned EEG inclusion from whole sessions to channel by epoch with sample-level bad-masking, roughly doubling usable data without loosening quality standards.

INSERM Lyon / CRNL **France**
Signal Processing Researcher (Master's Thesis) *Sept 2024–Oct 2025*

- Took a published decoding algorithm from paper through implementation to validation on real longitudinal recordings from a clinical cohort it had not been tested on (tetraplegia, locked-in syndrome, disorders of consciousness). Built the multi-subject pipeline: time-frequency decomposition, feature engineering over waveform components, PCA, discriminant classification, stratified cross-validation. Benchmarked against the conventional established method: AUC 0.72 against 0.66 in controls. It did not transfer to patients, reported as an honest null with the candidate explanations set out. Supervisors: Jérémie Mattout and James Bonaiuto.

Leibniz Institute for Resilience Research **Mainz, Germany**
Research Intern *May 2023–Jan 2024*

- Ported existing PsychoPy experimental paradigms to Java, preserving low-latency stimulus delivery and synchronised event marking.

Selected Projects

2026: **Sleep staging from wrist PPG** (in progress). Public DREAMT dataset, each person scored separately, feature groups added one at a time and kept only if they improve on the last.

2026: **Data-quality framework for a 111-subject public EEG corpus.** OpenNeuro `ds003775`, BIDS. The first QC pass flagged 57 to 64 of 64 channels in every subject; traced to raw DC offset, discarded and rerun on corrected data. Every preprocessing decision is written up in a public journal.

Technical Skills

Biomedical signals: EEG, ECG, PPG, respiration and inertial signals, high-frequency and longitudinal recordings. Heart rate and heart rate variability (SDNN, RMSSD, LF/HF), RR and inter-peak-interval analysis, respiratory rate, motion-artifact regression, signal-quality metrics.

Signal processing: Filtering (IIR and FIR), FFT and STFT, wavelets, superlet time-frequency decomposition, peak and transient detection, line-noise removal, artifact rejection, feature extraction.

Machine learning and validation: scikit-learn, PCA, discriminant and support-vector classification, convolutional networks for time-frequency classification (Keras, TensorFlow), feedforward networks in PyTorch, stratified and leave-one-recording-out cross-validation, mixed-effects models, permutation and resampling statistics, evaluation-harness design.

Data engineering: Python (NumPy, pandas, SciPy, MNE-Python, neurokit2, Matplotlib), MATLAB, Bash, Linux. PostgreSQL, TimescaleDB, PostgREST, FastAPI. Git, merge-request review, GitLab CI, pytest, Docker.

Education

2020–Oct 2025: **Integrated BS-MS (Master's), Biology major, Data Science minor.** IISER Mohali, India. NPTEL Silver Medalist (IIT Madras). Abstract accepted, Cortico BCI 2025.